📝 Digital SAT — Cheat Sheet

Procadamia | Strategies, Formulas & Quick Reference

TEST STRUCTURE & SCORING

Section	Questions	Time
Reading & Writing (2 modules)	54 questions	64 min
Math (2 modules)	44 questions	70 min
Total	**98 questions**	**~2h 14m**

Score	Approx. Percentile
1500–1600	Top 1–5%
1300–1490	Top 10–25%
1100–1290	50th–75th

> Adaptive: M2 difficulty set by M1 score. Strong M1 → harder M2 → higher ceiling. Maximize M1 correct answers.

UNIVERSAL STRATEGY RULES

- ⏱️ **Pace:** RW ~35 sec/Q | Math ~95 sec/Q
- 🚩 **Flag & skip** — mark hard Q, return within module
- 🧮 **Use Desmos** on every Math section
- ❌ **No wrong-answer penalty** — ALWAYS guess, never blank
- 🚫 **Cannot return** to a previous module after submitting
- ✅ **POE** — eliminate obviously wrong answers first

R&W WORDS IN CONTEXT

4-Step Strategy	
1. Cover	Blank out the word mentally
2. Context	Read sentence + surrounding lines
3. Predict	Form your own word before reading choices
4. Match	Pick the choice closest to your prediction

> ⚠️ Most common definition = most common WRONG answer. They always use the word in a more specific way.

Root	Key Word
sati-	**satiate** = fully satisfy
vener-	**venerate** = deeply respect
mitig-	**mitigate** = lessen severity
supplant	replace by force
temp-	**temperament** = natural disposition

R&W TEXT COMPLETION & PRECISION

CLUE → PREDICT → MATCH (pick the most precise fit)

Signal	Words	Predict
Contrast	however, but, despite, although	Opposite
Continue	in fact, moreover, additionally	Same direction
Cause-effect	because, therefore, thus	Logical result
Elaboration	for example, such as, including	More specific

> Rhetorical Synthesis: read stem → find the matching note → reject anything that adds outside info or contradicts notes.

R&W HARD COMPREHENSION QUESTIONS

Type	Key Rule
Command of Evidence (text)	Quote must DIRECTLY prove the claim. Related ≠ right.
Command of Evidence (data)	Find the exact data point. Close ≠ correct.
Inference	Must be true if text is true. No outside knowledge. Avoid extremes (always/never).
Text Structure/Purpose	Describe the FUNCTION (why), not the content (what).
Cross-text	Summarize each passage → identify agreement/disagreement → text only.

> Hard Q strategy: go back to passage FIRST, form your own answer, THEN read choices.

MATH ALGEBRA & LINEAR FUNCTIONS

Concept	Formula / Rule
Slope-intercept	$y = mx + b$
Slope from points	$m = (y_2-y_1)/(x_2-x_1)$
x-intercept	Set y = 0, solve for x
No solution	Same slopes, different y-intercepts (parallel)
Infinite solutions	Same slopes AND y-intercepts (identical lines)
One solution	Different slopes (lines intersect)
Absolute value	$\lvert A\rvert = c \rightarrow A = c$ OR $A = -c$

> No solution problems: set slopes equal, solve for the unknown parameter.

MATH RADICALS & RATIONAL EXPONENTS

$$a^{(m/n)} = \sqrt[n]{a^m}$$

Denominator = root index | Numerator = power

Expression	Value	Rule
$8^{(1/3)}$	2	$\sqrt[3]{8}$
$8^{(2/3)}$	4	$(\sqrt[3]{8})^2$
$27^{(1/3)}$	3	$\sqrt[3]{27}$
$16^{(3/4)}$	8	$(\sqrt[4]{16})^3$
a^{-n}	$1/a^n$	Negative exp.
$(a^m)^n$	a^{mn}	Power rule

MATH POLYNOMIALS, FACTORING & FUNCTIONS

Factoring Patterns	
Diff. of Squares	$a^2-b^2 = (a+b)(a-b)$
Perfect Square	$a^2 \pm 2ab+b^2 = (a \pm b)^2$
Trinomial (a=1)	Find p,q: p+q=b, pq=c

Factor Theorem

(x−c) is a factor ↔ f(c)=0. Test: plug in x=c, check if = 0.

Composite Functions f(g(x))	
Order	Inside out: evaluate g first, then f
Warning	f(g(x)) ≠ g(f(x))

Transform	Effect
$f(x+h)$	Shift LEFT h
$f(x-h)$	Shift RIGHT h
$f(x)+k$	Shift UP k
$-f(x)$	Reflect x-axis